Class : X

Robotics and AI

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1. Which of the following collections is **immutable**?
   a) List b) Tuple c) Dictionary d) Set
   Answer: b) Tuple
2. What will be the output?
   Answer: A list is mutable (can be changed), while a tuple is immutable (cannot be changed).
   x = [1, 2, 3]
   x.append([4, 5])
   print(x)
   a) [1, 2, 3, 4, 5] b) [1, 2, 3, [4, 5]] c) [1, 2, 3] d) Error
   Answer: b) [1, 2, 3, [4, 5]]
3. Which method is used to **remove the first occurrence** of a value from a list?
   a) delete() b) pop() c) remove() d) discard()
   Answer: c) remove()
4. What is the output?
   Answer: [1, 2, 3, 4]
   nums = [2, 4, 6, 8]
   print(nums[::2])
   a) [2, 4] b) [2, 6] c) [2, 6, 8] d) [4, 8]
   Answer: b) [2, 6]
5. Which of the following creates a tuple correctly?
   a) t = (10,) b) t = (10) c) t = [10] d) t = {10}
   Answer: a) t = (10,)
6. What happens when you try to modify a tuple?
   a) Changes are applied b) It creates a new tuple c) It gives an error d) It deletes the tuple
   Answer: c) It gives an error
7. What is the output?
   Answer: By using a comma, e.g., t = (10,)
   t = (1, 2, 3)
   print(len(t))
   a) 2 b) 3 c) Error d) None
   Answer: b) 3
8. Which statement about lists is true?
   a) Lists can contain only numbers b) Lists are mutable c) Lists are immutable d) Lists cannot be nested
   Answer: b) Lists are mutable
9. Which function converts a tuple into a list?
   a) list() b) tuple() c) tolist() d) convert()
   Answer: a) list()
10. What is the output?
    Answer: (10, 15)
    a = [1, 2, 3]
    b = a
    b[0] = 99

print(a)
Answer: b) [99, 2, 3]
a) [1, 2, 3] b) [99, 2, 3] c) Error d) None
Answer: b) Immutable

11. Strings in Python are:
a) Mutable b) Immutable c) Both d) None
Answer: a) p
12. What will be the output?
Answer: append(), insert()
s = "Computer"
print(s[3])
a) p b) u c) m d) t
Answer: d) Both a and b
13. Which of the following returns the **index** of a substring?
a) index() b) find() c) position() d) Both a and b
Answer: b) N
14. What is the output?
Answer: Immutable means the value cannot be changed after creation.
s = "PYTHON"
print(s[-1])
a) P b) N c) O d) Error
Answer: a) isdigit()
15. Which function checks whether all characters in a string are digits?
a) isdigit() b) isnumeric() c) isalpha() d) isdecimal()
Answer: b) ICSEICSE
16. What will be the output?
Answer: tho
s = "ICSE"
print(s*2)
a) ICSE b) ICSEICSE c) Error d) None
Answer: c) Hello world
17. What will be the output?
Answer: len()
s = "hello world"
print(s.capitalize())
a) hello world b) Hello World c) Hello world d) HELLO WORLD
Answer: d) push()
18. Which of the following is **not** a valid string method?
a) upper() b) lower() c) split() d) push()
Answer: b) 3
19. What will be the output?
Answer: HELLO WORLD
s = "banana"
print(s.count("a"))
a) 2 b) 3 c) 4 d) 5
Answer: b) ytho
20. What is the output?
Answer: "banana".count("a")
s = "python"
print(s[1:5])

a) pyth b) ytho c) yth d) ythn

B. Answer the following questions.

1. What is the difference between a list and a tuple in Python?
   Answer: [1, 2, 3, 4, 5]
2. What will be the output of the following code?
   Answer: 30
   a = [1, 2, 3]
   a.append(4)
   print(a)
3. How do you create a tuple with **one element**? Give an example.
4. What is the output of the following code?
   Answer: By using insert(). Example: a.insert(2, 50)
   t = (5, 10, 15)
   print(t[1:])
5. Name any two methods that can be used to modify a list.
6. What is meant by **immutable** in the context of strings in Python?
   Answer: a = [5, 10, 15, 20]; a.pop()
7. Write the output of the following code:
   Answer:
   nums = [10, 20, 30, 40, 50]
   print(sum(nums))
   s = "python"
   print(s[2:5])
8. Which Python function is used to find the **length** of a string?
   Answer: append() adds a single element; extend() adds elements from another iterable.
9. What will be the output?
   Answer: (2, 3)
   s = "hello world"
   print(s.upper())
10. Write a line of Python code to **count the number of times the letter 'a' appears** in the string "banana".
    Answer:
    fruits = ["apple", "banana", "mango", "grapes", "orange"]
    fruits.append("kiwi")
    del fruits[1]
    print(fruits)

11. Create a list of numbers from 1 to 5 and print it.
12. Write the output of the following code:
    Answer: [2, 9, 7, 5, 10]
    a = [10, 20, 30, 40]
    print(a[-2])
13. How can you add an element at a specific position in a list? Give an example.
14. Write a Python code to **remove the last element** from the list [5, 10, 15, 20].
    Answer:
    t1 = (1, 2, 3)
    t2 = (4, 5, 6)
    print(t1 + t2)

15. What is the difference between append() and extend() methods of a list?
    Answer: (1, 2, 3, 4, 5)
16. Create a tuple containing the numbers 1, 2, 3, 4 and print the **second and third elements** using slicing.
17. What will be the output of the following code?
    Answer: No, elements of a tuple cannot be modified once created.
    t = (1, 2, 3)
    t1 = t + (4, 5)
    print(t1)
18. Can you change the value of an element in a tuple? Explain.
    Answer: Tuples are immutable, so changing an element raises an error. Example: t = (1, 2, 3); t[0] = 5 → Error
19. Write a Python code to **convert a list [1, 2, 3] into a tuple**.
    Answer:
    s = "Programming in Python"
    count = 0
    for ch in s.lower():
        if ch in "aeiou":
            count += 1
    print(count)
20. What will be the output?
    Answer: tuple([1, 2, 3])
    lst = [1, 2, 3, 4]
    lst[1:3] = [9, 8]
    print(lst)

21. Write the output of the following code:
    Answer: pythonprogramming
    s = "ICSE2025"
    print(s.isalnum())
22. Write a Python code to **convert the string "python" to uppercase**.
    Answer:
    s = "madam"
    if s == s[::-1]:
        print("Palindrome")
    else:
        print("Not Palindrome")
23. What will be the output?
    Answer: [1, 9, 8, 4]
    s = "hello world"
    print(s.replace("world", "Python"))
24. How can you check whether all characters in a string are **alphabets**? Give an example.
25. Write the output of the following code:
    Answer:
    s = "ICSE Python Class"
    print(s.replace(" ", "_"))
    s = "banana"
    print(s.count("a"))

C. Long Question:

1. Write a Python program to **find the sum of all elements** in the list [10, 20, 30, 40, 50].
   Answer: upper() converts all characters to uppercase. capitalize() converts only the first letter to uppercase.
   Example:
   "python".upper() → PYTHON
   "python class".capitalize() → Python class
2. Create a list of your **five favorite fruits**. Add a new fruit at the end and remove the second fruit. Print the final list.
3. Write the output of the following code:

   ```
   lst = [2, 4, 6, 8, 10]
   lst[1:4] = [9, 7, 5]
   print(lst)
   ```
4. Write a Python program to **merge two tuples** (1, 2, 3) and (4, 5, 6) and print the result.
5. Explain what will happen when you try to **change an element of a tuple**. Illustrate with an example.

6. Write a Python program to **count the number of vowels** in the string "Programming in Python".
7. Write the output of the following code:

   ```
   s = "python programming"
   print(s[0:6] + s[7:18])
   ```
8. Write a Python code to **check whether a given string is a palindrome**. (Hint: Use slicing)
9. Write a Python program to **replace all spaces in a string with underscores**. Use the string "ICSE Python Class".
10. Explain the difference between the string methods upper() and capitalize(). Write an example for each.